

Math 157 - Code Architecture (Final Project)

1. Brainstormed Repo Structure (ContractModel)
 - a. Main.lean
 - b. Basic.lean
 - c. Definitions.lean
 - d. Contracts.lean
 - e. Examples.lean
 - f. EdgeCases.lean
 - g. [Z3Notes.md](#)
 - h. [README.md](#)
2. File Summaries
 - a. Main.lean - main file
 - b. Basic.lean - simple theorem experiments and Lean syntax practice
 - c. Definitions.lean - Defines agreement, contract, offer, acceptance, and related logical predicates
 - d. Contracts.lean - has core contract validity theorems
 - e. Examples.lean - has valid and invalid contract cases
 - f. EdgeCases.lean - models acceptance (conditional) and varied ambiguity
 - g. [Z3Notes.md](#) - Documents SMT/Z3 comparison experiments - reflection for future use

Core Lean Structures and example theorems present in implementation.

3. Planned Edge Cases
 - a. Conditional acceptance
 - b. Missing consideration
 - c. Revoked offers
 - d. Ambiguous acceptance
 - e. Mutual assent (for future project expansion)
4. Future Project Expansion Ideas
 - a. Z3 Integration (fully)
 - b. Additional legal concepts and considerations
 - c. Lightweight automation tactics